USER BEHAVIOR & ADVERTISEMENT ANALYTICS DATASET

Version: 5-2-2025 Encoded
Last Updated: June 23, 2025
Total Records: 350,000+ across 9 data tables
Format: .CSV

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OVERVIEW

This comprehensive dataset contains large-scale user demographic information, behavioral patterns, privacy preferences, and detailed advertisement interaction analytics. The dataset is specifically designed for advanced research in digital marketing effectiveness, user privacy impact analysis, behavioral targeting optimization, and cross-platform advertising performance measurement.

The dataset represents a significant expansion from previous versions, incorporating real-time interaction tracking, enhanced privacy consent modeling, and comprehensive engagement metrics across multiple advertising channels.

DATASET COMPOSITION

The dataset consists of 9 interconnected tables with over 350,000 total records:

Table Name	Records	Dimensions	Primary Focus
Structure	27	27×10	Dataset metadata and schema
Coding_Table	21	21×14	Categorical variable encodings
UserTable	50,003	50,003×7	User demographics and profiles
UserBehaviorTable	50,002	50,002×5	Browsing and interaction patterns
Privacy-Related_Features	50,002	50,002×5	Privacy settings and consent
AdCharacteristicsTable	62	62×3	Advertisement content and targeting
User_Interaction_Table	300,002	300,002×4	User-advertisement interactions
interaction	62	62×19	Detailed engagement metrics
intercation_Per_AD_Table	1,004	1,004×2	Per-advertisement summaries

DETAILED TABLE DESCRIPTIONS

1. Structure Table

Purpose: Dataset documentation and metadata schema
Content: Data dictionary, field definitions, version information
Key Features: Complete documentation of encoding schemes and data relationships

2. Coding Table

Purpose: Categorical variable encoding and mapping reference
Fields: Gender codes, Race/Ethnicity mappings, Privacy level definitions
Content: Standardized encoding for all categorical variables across dataset
Key Features: Enables consistent interpretation of encoded values

3. UserTable (50,003 records)

Purpose: Comprehensive user demographic and profile information
Key Fields:

- **User_ID:** Unique identifier (Primary Key)
- **Gender:** Encoded gender classification (0=Male, 1=Female)
- **Race_Ethnicity:** Demographic classification with numerical encoding

Research Applications: User segmentation, demographic analysis, diversity studies

4. UserBehaviorTable (50,002 records)

Purpose: Detailed user behavioral tracking and interaction patterns
Key Fields:

- **ID:** Record identifier (Primary Key)
- **User_ID:** Foreign key linking to UserTable
- **Browsing_History:** Complete URL navigation patterns and web activity
- **Interaction_Data:** Technical metadata including user agents and device info
- **Purchase_History:** Transaction records and product interaction data

Research Applications: Behavioral modeling, pattern recognition, predictive analytics

5. Privacy-Related Features (50,002 records)

Purpose: User privacy preferences and data sharing consent analysis
Key Fields:

- **ID:** Record identifier (Primary Key)
- **User_ID:** Foreign key linking to UserTable

- **Data_Sharing_Permissions:** Hierarchical privacy levels (encoded 1-4)
- **Privacy_Settings:** Category-based privacy preferences (encoded 20-30)
- **User_Consent:** Consent status and agreement levels (encoded values)

Research Applications: Privacy impact studies, GDPR compliance analysis, consent modeling

6. AdCharacteristicsTable (62 advertisements)

Purpose: Advertisement content analysis and targeting specification
Key Fields:

- **Ad_ID:** Unique advertisement identifier (Primary Key)
- **Ad_Content:** Complete advertisement text and messaging content
- **Ad_Targeting_Criteria:** Demographic and geographic targeting parameters

Research Applications: Content effectiveness analysis, targeting optimization

7. User Interaction Table (300,002 interactions)

Purpose: Comprehensive user-advertisement interaction tracking
Key Fields:

- **ID:** Interaction record identifier (Primary Key)
- **User_ID:** Foreign key linking to UserTable
- **Ad_ID:** Foreign key linking to AdCharacteristicsTable
- **Interaction_Type:** Encoded interaction classification (30+ categories)

Research Applications: Engagement analysis, click-through modeling, conversion tracking

8. Interaction Metrics Table (62 records)

Purpose: Detailed advertisement performance and engagement analytics
Key Fields:

- **ID:** Record identifier (Primary Key)
- **Ad_ID:** Foreign key linking to AdCharacteristicsTable
- **Likes, Shares, Comments:** Social engagement metrics
- **Clicks, Impressions:** Core performance indicators
- **Conversions:** Successful conversion events
- **Total_Interaction:** Aggregate engagement measure
- **Interaction_Ratio:** Engagement rate calculations
- **Conversion_Rate_Per_Impressions:** Performance efficiency metrics
- **Click_Rate_Per_Impressions:** Click-through rate calculations
- **Weighted_Sum:** Composite performance score
- **Engagement_Score:** Overall advertisement effectiveness rating

Research Applications: ROI analysis, performance optimization, A/B testing evaluation

9. Interaction Per AD Summary Table (1,004 records)

Purpose: Aggregated per-advertisement interaction summaries
Key Fields:

- **AdID:** Advertisement identifier
- **Aggregated metrics and summary statistics**

Research Applications: Campaign performance comparison, trend analysis

DATA ENCODING AND TECHNICAL SPECIFICATIONS

Categorical Variable Encoding

- **Gender:** 0=Male, 1=Female
- **Race_Ethnicity:** 10-18 (Native Hawaiian, American Indian, etc.)
- **Data_Sharing_Permissions:** 1-4 (level1 through level4)
- **Privacy_Settings:** 20-30 (Sales, Services, etc.)
- **Interaction_Types:** 30+ encoded interaction categories

Data Quality Characteristics

- **Completeness:** >99% data completeness across all primary fields
- **Consistency:** Standardized encoding schemes applied uniformly
- **Integrity:** Referential integrity is maintained through foreign key relationships
- **Scale:** Large-scale dataset suitable for machine learning and statistical modeling
- **Anonymization:** All PII anonymized while maintaining analytical utility

Technical Format Specifications

- **Character Encoding:** UTF-8 compatible for international character support
- **Numerical Precision:** Floating-point values maintained to 6 decimal places
- **Missing Values:** Represented as empty strings in CSV exports
- **File Format:** Comma-separated values (.csv) for universal compatibility
- **Chunk Processing:** Large tables processed in 10,000-row chunks for memory efficiency

RESEARCH APPLICATIONS AND USE CASES

Primary Research Domains

1. **Digital Marketing Effectiveness:** Large-scale analysis of advertisement targeting accuracy, conversion optimization, and cross-platform performance measurement

2. **Privacy Impact Assessment:** Comprehensive correlation analysis between privacy preferences, consent levels, and user engagement patterns across different demographic segments
3. **Behavioral Analytics and Modeling:** Advanced user segmentation based on browsing patterns, purchase behaviors, and interaction preferences using machine learning techniques
4. **Cross-Platform Advertising Analysis:** Multi-dimensional analysis of user interactions across different digital touchpoints and advertising channels
5. **Engagement Optimization:** Detailed performance analysis for advertisement content optimization and targeting strategy refinement

Advanced Analytics Capabilities

- **Machine Learning Applications:** Sufficient scale for deep learning models, recommendation systems, and predictive analytics
- **Statistical Modeling:** Robust sample size (50,000+ users) enables sophisticated statistical analysis and hypothesis testing
- **Time Series Analysis:** Interaction data supports temporal pattern analysis and trend identification
- **Network Analysis:** User-advertisement interaction patterns enable social network and influence analysis
- **Experimental Design:** The Dataset structure supports A/B testing, multivariate analysis, and controlled experiment evaluation

Data Loading Examples

Basic data loading:

```
import pandas as pd

# Load core tables
users = pd.read_csv('UserTable.csv')
behavior = pd.read_csv('UserBehaviorTable.csv')
privacy = pd.read_csv('Privacy-Related_Features.csv')
interactions = pd.read_csv('User_Interaction_Table.csv')

# Load reference tables
coding = pd.read_csv('Coding_Table.csv')
ads = pd.read_csv('AdCharacteristicsTable.csv')
```

Memory-efficient loading for large tables:

```
# Process large interaction table in chunks
chunk_size = 10000
interaction_chunks = pd.read_csv('User_Interaction_Table.csv', chunksize=chunk_size)

for chunk in interaction_chunks:
    # Process each chunk individually
    processed_chunk = process_data(chunk)
```

Advanced analysis setup:

```
# Merge tables for comprehensive analysis
user_profile = users.merge(behavior, on='User_ID').merge(privacy, on='User_ID')

# Create interaction analysis dataset
full_interactions = interactions.merge(users, on='User_ID').merge(ads, on='Ad_ID')
```

DATA GOVERNANCE AND ETHICS

Privacy and Compliance

- **Data Anonymization:** All personally identifiable information (PII) has been completely anonymized using industry-standard techniques while preserving analytical utility
- **Consent Framework:** User consent protocols are rigorously followed for all data collection activities in compliance with international privacy regulations
- **GDPR Compliance:** The Dataset structure and content are fully compliant with the General Data Protection Regulation requirements for research purposes
- **Ethical Research Standards:** Data collection and processing adhere to established ethical frameworks for human subjects research in digital environments
- **Research Integrity:** Transparent methodology and data provenance documentation are maintained throughout the collection and processing phases

Academic Usage Guidelines

- **Citation Requirements:** Academic publications using this dataset must include appropriate citation and acknowledge data source contributions
- **IRB Compliance:** Researchers must ensure compliance with institutional review board requirements for human subjects research before dataset utilization
- **Publication Standards:** Results derived from this dataset should adhere to established standards for reproducible research and open science practices
- **Data Sharing:** Derivative datasets and analysis results should follow appropriate data sharing protocols while maintaining privacy protections

Limitations and Considerations

- **Temporal Scope:** The Dataset represents a cross-sectional snapshot with limited temporal coverage for longitudinal analysis
- **Geographic Bias:** Potential geographic and cultural biases present in the user sample may limit generalizability across different populations
- **Platform Specificity:** Behavioral data is subject to platform-specific limitations and measurement constraints
- **Interaction Tracking:** Advertisement performance metrics are limited to tracked interactions and may not capture all user engagement forms
- **Sample Representativeness:** While a large-scale sample may not be fully representative of the broader digital advertising ecosystem